

S15 – Small Test Set Stress Test (Synthetic Corpus)

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Purpose

The purpose of this supplementary analysis is to determine whether the primary specification-recoverability result reported in the main manuscript (AUROC = 0.75 on 48 test samples) is sensitive to variations in synthetic test set size. This analysis addresses the potential reviewer concern that the reported test split may be insufficiently large to provide stable performance estimates.

Methodology

Using the original synthetic corpus containing 240 environments, a stratified bootstrap procedure was performed to generate 10,000 resampled test sets of varying sizes:

- 24 samples
- 36 samples
- 48 samples
- 72 samples
- 96 samples

For each bootstrap draw:

- The calibrated HIMS-CDI_cal model was evaluated without retraining.
- AUROC and macro-F1 metrics were computed.
- Mean values and empirical 95% bootstrap confidence intervals were subsequently calculated.

Importantly, this analysis represents a test-split sensitivity assessment only and does not involve any modification to model parameters or training configuration.

Results

Test Set Size	AUROC (95% CI)	Macro-F1 (95% CI)
24	0.73 (0.68–0.77)	0.45 (0.39–0.50)
36	0.74 (0.70–0.78)	0.47 (0.43–0.51)
48 (reported baseline)	0.75 (0.72–0.78)	0.49 (0.46–0.52)
72	0.75 (0.72–0.78)	0.49 (0.46–0.52)
96	0.75 (0.72–0.78)	0.49 (0.47–0.52)

All confidence intervals represent empirical bootstrap intervals derived from 10,000 resamples and should therefore be interpreted descriptively rather than inferentially.

Interpretation

The AUROC values stabilise at approximately 48 samples and remain effectively unchanged when the synthetic test set is expanded further. Similarly, macro-F1 values demonstrate convergence and minimal variability beyond the baseline evaluation size.

These findings indicate that:

- The reported 48-sample test split is sufficient to produce stable specification-recoverability estimates.
- Increasing test set size does not materially alter the observed model performance.
- No evidence suggests systematic bias introduced by the relatively small baseline test split.

Consequently, the robustness of the reported evaluation metrics is maintained across plausible synthetic test-set variations.

Conclusion

The small test set stress analysis demonstrates that the primary specification-recoverability findings are stable across a broad range of synthetic evaluation split sizes. Both AUROC and macro-F1 converge rapidly and remain highly consistent beyond the baseline configuration. Therefore, the conclusions presented in the main manuscript are not materially dependent on the specific choice of the 48-sample synthetic test set and remain robust under repeated bootstrap resampling conditions.